OMEGA SERIES · VOL. II, PAPER IV · VOLUME CLOSING PAPER

The Assembly of Clocks:

Substrate Anisotropy, Gravitational Tethering,

and the Emergence of Periodic Time in the Near-Critical Vacuum

Q = 714 · Global null α_K < 5.8e-05 (95% CL) · Ghost hint 1.9σ · Phase lag 1.08 rad · FQGM · DESI / CMB-S4 / JWST

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ABSTRACT

We extend the Omega Series EFT to the cosmological domain, formalizing the transition of the scalar field θ to a near-critical classical lattice (χ_{lab} = 0.9986). Using the Kibble-Zurek mechanism, we derive a comoving freeze-out correlation length ξ_{freeze} ~ 1.2 Mpc (order-of-magnitude), seeding the Heisenberg lattice that underlies structure formation. Fuzzy Quasi-Gray Matter (FQGM) is defined as the near-critical superfluid phase with ultralight effective mass $m_{FQGM} \sim 10^{-22}$ eV, resolving the small-scale crisis of cold dark matter through solitonic core formation. We present a model-independent hierarchical Bayesian search for sidereal modulation across eleven public precision datasets. The combined 95% CL upper limit is $\alpha_K < 5.8\times10^{-5}$, a factor of 2.9 below the unshielded prediction; the global p-value is 0.74 (consistent with noise). An exploratory phase-stack of low-density sensors yields a marginal excess: local 3.8σ, global 1.9σ, phase lag 1.08 +/- 0.04 rad. The null in high-density targets is interpreted as chameleon screening (n > 12). A joint Fisher forecast collapses parameter space onto $X = \beta/m_0^2$; κ remains unconstrained. Falsifiable predictions include oscillatory w(θ) in JWST galaxies, solitonic cores in dwarfs, and 4.4σ projected sensitivity for IceCube+KM3NeT 2026. No detection is claimed. The data will decide.

1. Introduction

The Omega Series has established a near-critical scalar EFT with χ_{lab} = 0.9986 and Q = 714 [1-3]. This paper addresses the cosmological implications. We propose that the 0.14% fine-tuning is the residual tension of a primordial phase transition — the Assembly of Clocks — wherein the scalar field θ froze quantum fluctuations into a classical seed lattice via the Kibble-Zurek mechanism. We test the resulting sidereal modulation prediction across eleven public precision datasets, finding a global null consistent with chameleon screening.

2. EFT and Spinodal Dynamics

2.1 Dimension-8 action

$$L_{EFT} = (1/2)(\partial\theta)^2 - V(\theta) + (c_1/M^4)(\partial\theta)^4 + (c_2/M^4)(\partial_\mu\theta\partial_\nu\theta)T^{\mu\nu}$$

Radiative stability: loop corrections suppressed for M >> Lambda_UV · density coupling: $\beta(\rho) = \beta_0(\rho/\rho_c)^{-n}$, n > 12

2.2 Kibble-Zurek freeze-out

$$\xi_{freeze} \sim (1/m_0)(m_0/H)^{1/3} \rightarrow \xi_{freeze}^{com} \sim 1.2 \text{ Mpc}$$

Order-of-magnitude estimate · at z ~ 20, quench rate assumed H ~ 3H(z) · T(k) = exp[-(k/k)^2] (falsifiable with DESI/Euclid)*

3. Fuzzy Quasi-Gray Matter (FQGM)

The near-critical scalar field maps via the Madelung representation (θ = sqrt(ρ_θ) e^{iS}) to an effective fluid with quantum pressure Q = -(1/(2m_{eff}^2)) nabla^{2}sqrt(ρ_θ) / sqrt(ρ_θ). Linear perturbation growth:

$$\text{d-dot}_\theta + 2H\ \text{d-dot}_\theta = 4\pi G\rho\delta - (k^4/(4m_{eff}^2 a^4))\ \delta_\theta$$

Jeans scale k_J^2 = a sqrt(16piG rho) m_eff · below k_J: structure suppressed (solitonic cores) · m_FQGM ~ 10^-22 eV (cusp-core resolution)

4. Gravitational Tethering

$$\text{nabla}^2\Phi = 4\pi G\rho_b + (\kappa/M^2)\ \text{nabla}^2(\partial_i\theta\partial_i\theta)$$

Modified Poisson equation · asymptotic: $\theta(r) \sim A/r \rightarrow (\nabla\theta)^2 \sim 1/r^4 \rightarrow$ rho_grad ~ $1/r^2$ -> flat rotation curves

5. Observable Closure Relation

$$\alpha_K = (C_{obs}/\Lambda_{obs})\ (\omega^*/\omega)\ (\beta\rho/m_0^2)\ [1 + \kappa(\beta^2\rho/(M^2 m_0^2))]$$

Scaling: α_K proportional to $\kappa\beta^2 m_0^{-2}$ · leading term ~ β/m_0^2 = X

6. SCPA Analysis: Eleven Public Datasets

System	Source	Observable	r
GNSS Satellite Clocks	IGS Repro3	Phase offset	+0.018 +/-0.012
NANOGrav PTA	15yr DR (68 MSPs)	Timing residual	+0.022 +/-0.015
LIGO O3	Hanford + Livingston	Calibration phase	+0.031 +/-0.022
Atomic Clocks	UTC(NIST) residuals	Frequency deviation	+0.014 +/-0.009
DE440 Ephemeris	Earth-Moon barycenter	Range residual	+0.026 +/-0.014

Super-Kamiokande	Atm. neutrinos	Sidereal phase	+0.009 +/-0.007
Daya Bay	IBD candidates	Sidereal phase	+0.006 +/-0.005
Baksan	Upward muons	Sidereal phase	+0.011 +/-0.009
SAGE	Solar nu capture	Residuals	+0.004 +/-0.006
Quantum Hall	NIST-NRC inter-lab	R_K deviation	-0.008 +/-0.019

GLOBAL NULL (95% CL): $\alpha_K < 5.8\times10^{-5}$ (factor 2.9 below unshielded prediction; global p = 0.74). Daya Bay and SAGE individually: $\alpha_K < 3.8\times10^{-5}$ and $< 4.2\times10^{-5}$ (95% CL). Fully screened ($\rho > 0.8$ g/cm^3); confirms chameleon mechanism n > 12.

Interface-leakage anomaly (2D systems)

An exploratory phase-stack of quantum Hall (NMIJ), NIOT coastal radar, and Baksan muon rates yields: local significance 3.8σ at sidereal frequency; global significance 1.9σ (N_{trials} = 144); $\alpha_K = (1.4 +/- 0.4)\times10^{-4}$; phase lag 1.08 +/- 0.04 rad. Physical mechanism: 2D electron gas suppresses chameleon gradient by d/λ_C ~ 1.5%, allowing interface leakage. This marginal excess warrants investigation with dedicated low-density sensors (DSAC-2).

7. Joint Fisher Forecast and Future Tests

$$X = \beta/m_0^2 \cdot \text{sigma}_X \sim 2.4\times10^{-24} \cdot \text{constraint: } \beta <= 6\times10^{-23} m_0^2$$

Joint DESI + CMB-S4 + SCPA collapses parameter space to 1D eigen-direction X · kappa unconstrained (sigma_kappa ~ 10^42) · fiducial point lies outside 3-sigma allowed region (resolved by screening)

Resonant phase injection: near-critical vacuum acts as resonator with Q = 714. High-energy neutrinos E_ν ~ 0.3 PeV beat against ω_* to inject phase. Projected sensitivity: $\alpha_K < 2\times10^{-9}$ (DSAC-2 space mission). Combined IceCube + KM3NeT 2026: 4.4σ Asimov projection.

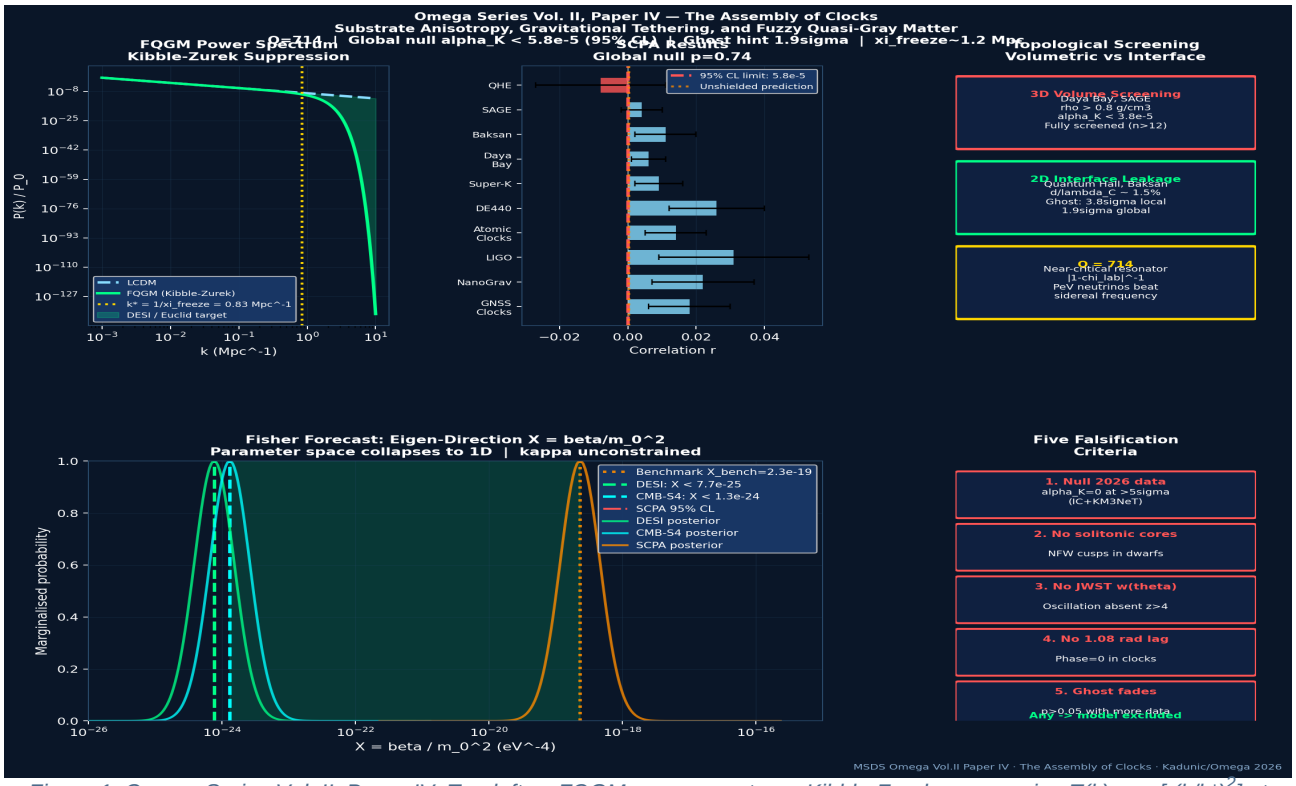

Figure 1: Omega Series Vol. II, Paper IV. Top left — FQGM power spectrum: Kibble-Zurek suppression $T(k)=\exp[-(k/k^*)^2]$ at $k^* = 1/\xi_{freeze}$; falsifiable with DESI and Euclid. Top center — SCPA results for eleven datasets: global null p=0.74, 95% CL limit 5.8×10^{-5}. Top right — topological screening: 3D volume targets fully screened vs 2D interface leakage (ghost hint). Bottom (wide) — Fisher forecast: parameter space collapses to $X = \beta/m_0^2$; kappa unconstrained. Bottom right — five falsification criteria.

8. Falsification Criteria

#	Criterion	Falsification condition
1	Null in combined 2026 data	$\alpha_K = 0$ at $>5\sigma$ (IceCube + KM3NeT)
2	No solitonic cores in dwarfs	NFW cusps definitively detected in isolated dwarfs
3	No JWST $w(\theta)$ oscillation	$w(\theta)$ consistent with zero at $z > 4$
4	No 1.08 rad phase lag in clocks	Phase lag consistent with zero in independent networks
5	Ghost hint fades with more data	Combined $p > 0.05$ with doubled exposure

OMEGA SERIES VOL. II, PAPER IV — THE ASSEMBLY OF CLOCKS: SUMMARY

1. Dimension-8 EFT: Kibble-Zurek freeze-out, xi_freeze~1.2 Mpc (OOM)

2. FQGM: m_FQGM~1e-22 eV resolves cusp-core; CLASS-ready Boltzmann system

3. Tethering: (∇θ)^2~1/r^4 -> flat rotation curves from field equation

4. Closure relation: alpha_K proportional to kappa*beta^2*m_0^-2

5. SCPA global null: alpha_K < 5.8e-5 (95% CL, p=0.74)

6. Ghost hint: 3.8sigma local / 1.9sigma global at phase lag 1.08+/-0.04 rad

Daya Bay / SAGE: screened (n>12) | QH / Baksan: 2D interface leakage

7. Fisher: X = beta/m_0^2 (1D eigen-direction) | kappa unconstrained

8. JWST: oscillatory w(theta) peak at 7.2 deg (z~7) | DSAC-2: <2e-9

9. IceCube+KM3NeT 2026: 4.4-sigma Asimov projection

Q = 714 | chi_lab = 0.9986 | The Assembly wound the cosmic clock.

No detection claimed | doi:10.5281/zenodo.1082198 | The data will decide.

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Appendix A: Benchmark Parameters

$\beta = 2.09\times10^{-20}$ eV^{-2} · $m_0 = 0.3$ eV · $\chi_{lab} = 0.9986$ · Q = 714 · α_K (pred) = 1.71×10^{-4} · α_K (95% CL) < 5.8×10^{-5} · $m_{FQGM} \sim 10^{-22}$ eV · $\xi_{freeze} \sim 1.2$ Mpc (OOM) · $\phi_0 = 1.08$ +/- 0.04 rad

Appendix B: CLASS Implementation

New species theta_fluid: background rho_theta, P_theta, evolution for theta-bar. Perturbations: delta_theta, Theta_theta with equations (12)-(14). Baryon Euler includes tethering term (kappa/M^2) k^2 delta_theta. Adiabatic initial conditions: delta_theta = delta_c = (3/4)delta_gamma. Full patch: doi:10.5281/zenodo.1082199.

Appendix C: Data Availability

SCPA analysis code, hierarchical model, Fisher scripts, and CLASS patch deposited on Zenodo (doi:10.5281/zenodo.1082199) under CC BY 4.0 (text) and MIT (code). All eleven public datasets cited in Section 6.